

Multimodal Alpha Forecasting and Robust Portfolio Optimization

*15.095 Machine Learning Under a Modern
Optimization Lens*

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November 29, 2025

Abstract

.....To be completed.....

1 Introduction

Short-horizon equity return prediction remains one of the most challenging problems in quantitative finance. Daily returns are dominated by noise, exhibit low signal-to-noise ratios, heavily influenced by market microstructure dynamics, and reflect nonlinear interactions among systematic factors, macroeconomic conditions, and firm-specific characteristics. Despite this difficulty, even marginal improvements in predictive accuracy can yield economically meaningful gains when used within a risk-controlled portfolio optimization framework.

The goal of this project is to develop a multimodal forecasting system that integrates complementary sources of information: price-based technical indicators, firm fundamentals, macroeconomic variables, sector and subsector classifications, calendar-time structure, sentiment-derived features and text embeddings from news data. We aim to evaluate which feature groups contribute meaningful predictive power and generate a next-day return forecast suitable for prescriptive portfolio optimization.

2 Data

2.1 Data Sources

Because no single source provides all the information required for forecasting and portfolio construction, we integrate data from multiple independent providers. Our final dataset combines four major categories of information:

- **S&P 500 Constituents:** The list of S&P 500 tickers during our sample period serves as the base universe.
- **Stock Market Data:** Daily OHLCV (Open, High, Low, Close, Volume) data for all S&P 500 stocks from 2016–2023 are obtained from Yahoo Finance.
- **Company Fundamentals:** Firm-level financial metrics (e.g., sales growth, margins, ROA, ROE, leverage, liquidity ratios, valuation ratios) are sourced from WRDS/Compustat.
- **Macroeconomic Indicators:** Broader economic conditions are captured using FRED variables such as GDP growth, unemployment rates, and Treasury yields.
- **News Data:** Daily financial news headlines and summaries are collected from the FNSPID dataset.¹

2.2 Data Processing Pipeline

After gathering raw inputs, we construct a unified panel dataset at the (ticker, date) level. The main steps are:

¹<https://huggingface.co/datasets/Zihan1004/FNSPID>

1. **Cleaning and Alignment:** All data sources are standardized to daily frequency. We merge by ticker and date, forward-fill fundamentals to match market trading days.
2. **Return Construction:** Using daily closing prices, we compute the next-day stock return, which serves as the prediction target.
3. **Feature Engineering:** We generate:

- ***Price-Based Features and Technical Indicators*** Daily OHLCV data serve as the foundation for constructing short-horizon technical indicators that are commonly used in quantitative trading:

- Moving averages at several horizons (2-day, 3-day, 1-week, 2-week, 1-month, 1.5-month, 2-month)
- Exponential moving averages (EMA) with windows of 10, 20, and 50 days
- Rolling-window return features (e.g., 3-day, 7-day, and 21-day average returns) that capture short-term momentum and drift in recent price movements
- Volume-derived proxies, such as log volume

These variables constitute the price-signal feature group, which forms the baseline for many of the models evaluated.

- ***Fundamental Variables*** Forward-filled quarterly financial statement items to incorporate firm-level characteristics:

- Profitability metrics, such as return on assets (ROA), return on equity (ROE), and gross, operating, and net margins
- Growth metrics, including sales growth, asset growth, equity growth
- Valuation ratios, such as book-to-market, earnings yield, cash-flow yield, and sales yield
- Liquidity and leverage ratios, including current ratio, leverage, and cash holdings

These fundamentals capture cross-sectional variation that evolves more slowly but is often predictive of medium-term equity behavior.

- ***Macroeconomic Indicators*** Daily or monthly macro variables aligned to trading days:

- Market volatility via VIX
- Yield curve spreads (10Y–2Y, 10Y–3M)
- Federal Funds Rate
- Economic activity indicators, including industrial production, unemployment, and GDP

These variables embed systematic risk dynamics and macro-sensitive return drivers.

- ***Sector and Subsector Structure*** Firm-level structure is incorporated using GICS (Global Industry Classification Standard) classifications:

- One-hot sector indicators
- One-hot subsector indicators

These features encode economically meaningful groupings that help models recognize sector-specific return patterns and reduce cross-sectional confounding.

4. **Handling Missing Values:**

- *Forward-filling time-invariant* (e.g., fundamentals such as leverage, ROA, and margins). These variables are forward-filled *within each firm* so that only past information is used, which prevents look-ahead bias.
5. **Train–Validation–Test Split:** To avoid look-ahead bias, we split the dataset chronologically:
- 2016–2021 for training,
 - 2022 for validation,
 - 2023 for out-of-sample testing.

This pipeline produces a complete, aligned dataset suitable for both prediction modeling and portfolio backtesting.

3 Methodology

3.1 Multimodality

Text Features

To incorporate unstructured information alongside traditional financial signals, we extract two types of text-based features from daily news headlines and summaries using the pretrained FinBERT model:

- **Sentiment Scores.** A FinBERT sentiment classifier assigns each news item a positive, negative, or neutral label. For each stock and each day, we aggregate these predictions into summary measures such as mean sentiment, extremal sentiment, and news count.
- **Text Embeddings.** Using the base FinBERT model, we obtain 768-dimensional contextual embeddings for each headline and summary. We compute the daily embedding for each stock by averaging all embeddings associated with its news on that date.

These features capture narrative tone, linguistic signals, and information flow not present in standard price or fundamental data.

3.1.1 Dimension Reduction

Because daily embeddings are high-dimensional, we apply PCA to project them onto 64 components. This retains the majority of the variance while keeping the modeling pipeline computationally manageable for a large cross-section of stocks.

3.2 Stock Next Day Return Prediction

Evaluation Metric

Since the goal of this project is to forecast next-day equity returns in a noisy, low-signal environment, the evaluation framework must reflect the decision-making requirements of a trading system. Although all models generate real-valued return predictions, the primary operational question is whether the model correctly anticipates the *direction* of the next day’s movement, as this determines the trading position taken.

To align with this objective, we use **Directional Accuracy (DA)** as the central evaluation metric. DA measures the fraction of predictions whose sign matches the sign of the realized next-day return:

$$\text{DA} = \frac{1}{N} \sum_{i=1}^N \mathbf{1} \{ \text{sign}(\hat{r}_i) = \text{sign}(r_i) \},$$

where $\hat{r}_i$ is the predicted next-day return, r_i is the realized next-day return, and $\mathbf{1}\{\cdot\}$ is the indicator function. A DA of 50% corresponds to random guessing in a balanced up/down environment, so improvements of even 1–2 percentage points are economically meaningful in this setting.

3.2.1 Baseline Models

Figure 1: **Top 5 baseline models ranked by train+validation accuracy.** The blue bars show training and validation directional accuracy, while the orange bars show out-of-sample test performance for 2023.

3.2.2 Hybrid Baselines

3.2.3 Linear Regression

3.2.4 Linear Regression

3.2.5 XGBoost

3.2.6 CatBoost

3.2.7 LightGBM

3.2.8 LSTM

3.2.9 TCN

3.3 Robust Portfolio Optimization

4 Results

compare the best model from each category traintest DA compare the return of each

5 Discussion

6 Conclusion

Appendix